

Kalyani Patrikar

🌐 kalyanipatrikar.github.io

📍 Liverpool, UK | 📞 +447818912702 | ✉ k.patrikar@liverpool.ac.uk

🐙 github.com/kalyanipatrikar | [in linkedin.com/in/kalyani-patrikar](https://www.linkedin.com/in/kalyani-patrikar)

Building multiscale computational frameworks with quantum chemical methods, physics-based models, and neural networks, to uncover mechanisms, enhance system performance, and design new materials

EXPERIENCE

Postdoctoral Research Associate, *University of Liverpool*

PRESENT

- Developed **physics-based model** integrating molecular dynamics data and charge transport theory for materials with time variant disorder; **exceeding** state-of-the-art mobility prediction **accuracy** for OMIECs
- Developing **algorithm** simulating charge transport in polymers based on asymmetric charge transfer rates

Early Career Fellow, *Indian Institute of Technology Gandhinagar*

Aug 2022 – Oct 2024

One of the highest postdoctoral level fellowships offered by any Indian Institute

- Developed a **multi-scale** framework (ab initio computation, semi empirical QM, kinetic Monte Carlo) to predict **system-level** behaviour from **first-principles** bypassing experimental parameters, validated with 14 OPV
- Fine tuned MACE-OFF** (sampling, labelling, training, validation) on AQx-based configurations to enable rapid conformer prediction across the AQx molecular family; **integrated** MLIP derived structures for downstream **charge transport** simulation pipeline
- Designed **graph neural network** based **surrogates** for quantum mechanical properties; achieved high accuracy for electronic as well as optical properties with attention networks by mixing data pooling methods
- Designed and built computational pipeline parsing 257 molecules into classified substructures for combinatorial generative algorithm, 500 000 **novel molecules** analysed by MPNN with DFT validation
- Developed **DFT and physics-based model** for mechanism at heterogeneous interfaces predicting **experimentally** observed behaviour of organic devices; extended model to apply for catalysts

Senior Research Assistant, *Center for Excellence in Nanoelectronics*

Jul 2013– Dec 2014

- Designed and fabricated oxide nanoparticles embedded organic transistors for toxic gas sensors

SKILLS

Programming : Python, Bash (& SLURM), C

Python Libraries : PyTorch (&PyG), MACE, ASE, RDKit, PySCF, NumPy, SciPy, Pandas

Software : CP2K, Gaussian16, Quantum Espresso, xTB, Jmol, VESTA, Sentaurus TCAD

AI Models : Graph Neural Networks (GAT, MPNN), CNN, Random Forest

KEY PUBLICATIONS (COMPLETE LIST AT kalyanipatrikar.github.io/publications)

Journal (*Equal Contribution Authors, † Corresponding Author)

- Kalyani Patrikar, Keval Patadia, Rudranarayan Khatua, Anirban Mondal, "From Molecules to Devices: A Multiscale Approach to Evaluating Organic Photovoltaics" *J. Chem. Theory Comput.*, 20, 22, 2024 [[10.1021/acs.jctc.4c01029](https://doi.org/10.1021/acs.jctc.4c01029)] < **Multiscale simulation pipeline DFT → xTB → kMC mapping atomistic physics to device behaviour** >
- Dinesh Trivedi*, Kalyani Patrikar*, Anirban Mondal, "Dataset Size Invariant Graph Attention Network for Prediction of Diverse Molecular Properties", *Mol. Syst. Des. Eng.*, 9, 2024 [[10.1039/D4ME00113C](https://doi.org/10.1039/D4ME00113C)] < **Versatile graph neural network surrogates for expensive quantum-mechanical materials properties** >

3. Kalyani Patrikar[†], Valipe Ramgopal Rao, Dinesh Kabra, Anirban Mondal[†], "Understanding the Microscopic Origin of the Contact Resistance at the Polymer–Electrode Interface", *ACS Appl. Mater. Interfaces*, 15 (42), 49427, 2023 [10.1021/acsami.3c10260] < **Physics-based model of interface charge transfer based on ab initio computations, explaining interface resistance trends observed in device measurements** >
4. Bibhas Das, Kalyani Patrikar, Atharva Keny, Anirban Mondal, "From fragments to function: data-driven design of high-performance non-fullerene acceptors for organic photovoltaics" *Mol. Syst. Des. Eng.*, 11, 269-291, 2026 [10.1039/D5ME00182J] < **End-to-end computational pipeline for combinatorial generative design of high-performance materials** >

Conference (*Presenting Author)

1. K. Patrikar*, C. Burke, A. Troisi, "Charge Transport Enhancement with Dynamic Disorder", Oral presentation at *European Conference in Molecular Electronics 2025* < **Physics-based model for charge transport in dynamically disordered system** > Awarded **Best Contributed Talk**

EDUCATION

Doctor of Philosophy

Indian Institute of Technology Bombay

Electrical Engineering

Jan 2015 – Dec 2021

Master of Technology

Indian Institute of Technology Bombay

Materials Science

Jul 2011 – Jun 2013

Bachelor of Technology

Visvesvaraya National Institute of Technology

Metallurgical Engineering & Materials Science

Jul 2007 – May 2011

PHD THESIS

Bulk and Interfacial Charge Transport in Organic Field Effect Transistors

- Demonstrated **novel** charge doping **mechanism** of self-assembled monolayers at transistor contacts; developed **device model** based on **ab initio computations** and device characterisation
- Established new molecular design paradigm for order-of-magnitude **device performance** improvement based on device characterisation, **DFT computations**, and **polaron transport** analysis; reported **highest mobility** for transistors of DA polymers PTB7 and PTB7Th
- Developed **multiscale model** simulating measured cryogenic device characteristics of organic transistors

LEADERSHIP AND COMMUNICATION

- **Teaching:** Flexible Electronics (2 Credit Course) *IIT Gandhinagar*
- **Demonstrator:** Introduction to Spectroscopy, **Chemical Data Discovery and Design** *University of Liverpool*
- **Teaching Assistant:** Physics of Transistors, **Machine Learning**, Microelectronics Lab, **Communication Skills**, VLSI Technology, Thermodynamics *IIT Bombay*
- **Students Mentored:** BTech and MSc. students for thesis project work *IIT Gandhinagar*

EXTRACURRICULAR ACHIEVEMENTS

- Organised and anchored events as the Institute Executive Member in IIT Bombay Academic Council
- Completed high altitude treks in Himalayas, India